1. Tabela Verdade

Considere a proposição composta:

$$(P \wedge \neg Q) \rightarrow (Q \vee R)$$

Construa a tabela verdade completa para a proposição.

Valor: 2,0 pontos

P	Q	R	~Q	P∧~Q	Q∨R	P∧~Q → Q∨R
V	V	V	F	F	V	V
V	V	F	F	F	V	V
V	F	V	V	V	V	V
V	F	F	V	V	F	F
F	V	V	F	F	V	V
F	V	F	F	F	V	V
F	F	V	V	F	V	V
F	F	F	V	F	F	V

2. Operações com Matrizes

Dadas as matrizes:

$$D = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \\ 2 & 1 & 0 \end{pmatrix}, \quad E = \begin{pmatrix} 2 & 0 & 1 \\ -1 & 3 & 2 \\ 1 & -2 & 1 \end{pmatrix}, \quad F = \begin{pmatrix} 3 & 1 & 0 \\ -2 & 0 & 4 \\ 1 & 2 & -1 \end{pmatrix}$$

Calcule:

a) $2D + E - 3F$

b) $D \cdot E^T$

c) $E \cdot F$

Valor: 2,0 pontos cada item

a)

$$2 \times \begin{bmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \\ 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -4 & 6 \\ 0 & 8 & -2 \\ 4 & 2 & 0 \end{bmatrix} \qquad 3 \times \begin{bmatrix} 3 & 1 & 0 \\ -2 & 0 & 4 \\ 1 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 9 & 3 & 0 \\ -6 & 0 & 12 \\ 3 & 6 & -3 \end{bmatrix}$$

$$2D+E-3F = \begin{bmatrix} 2 & -4 & 6 \\ 0 & 8 & -2 \\ 4 & 2 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 1 \\ -1 & 3 & 2 \\ 1 & -2 & 1 \end{bmatrix} - \begin{bmatrix} 9 & 3 & 0 \\ -6 & 0 & 12 \\ 3 & 6 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 4 & -4 & 7 \\ -1 & 11 & 0 \\ 5 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 9 & 3 & 0 \\ -6 & 0 & 12 \\ 3 & 6 & -3 \end{bmatrix} = \begin{bmatrix} -5 & -7 & 7 \\ -7 & 11 & -12 \\ 2 & -6 & -2 \end{bmatrix}$$

b)

$$E^T = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & -2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$D \times E^T = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \\ 2 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & -2 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 2+3, & -1-6+6, & 1+4+3 \\ -1, & 12-2, & -8-1 \\ 4, & -2+3, & 2-2 \end{bmatrix} = \begin{bmatrix} 5 & -1 & 8 \\ -1 & 10 & -9 \\ 4 & 1 & 0 \end{bmatrix}$$

c)

$$\begin{bmatrix} 2 & 0 & 1 \\ -1 & 3 & 2 \\ 1 & -2 & 1 \end{bmatrix} \times \begin{bmatrix} 3 & 1 & 0 \\ -2 & 0 & 4 \\ 1 & 2 & -1 \end{bmatrix} = \begin{matrix} 6+1, & 2+2, & -1 \\ -3-6+2, & -1+4, & 12-2 \\ 3+4+1, & 1+2, & -8-1 \end{matrix}$$

$$= \begin{matrix} 7 & 4 & -1 \\ -7 & 3 & 10 \\ 8 & 3 & -9 \end{matrix}$$

3. Determinantes e Inversas

Com relação às matrizes do exercício anterior, calcule:

a) Determinante de E

b) Matriz inversa de D e de F, caso existam

a)

$$\begin{bmatrix} 2 & 0 & 1 \\ -1 & 3 & 2 \\ 1 & -2 & 1 \end{bmatrix} \begin{matrix} 2 & 0 \\ -1 & 3 \\ 1 & -2 \end{matrix}$$

$2 \times 3 \times 1 = 6$

$1 \times (-1) \times (-2) = 2$

$6 + 2 = 8$

$0 \times (-1) \times 1 = 0$

$2 \times 2 \times (-2) = -8$

$1 \times 3 \times 1 = 3$

$-8 + 3 = -5$

$8 - 5 = 3$

b)

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \\ 2 & 1 & 0 \end{bmatrix} \begin{matrix} 1 & -2 \\ 0 & 4 \\ 2 & 1 \end{matrix}$$

$1 \times 4 \times 0 = 0$

$-2 \times (-1) \times 2 = 4$

$3 \times 0 \times 1 = 0$

4

$-2 \times 0 \times 0 = 0$

$1 \times (-1) \times 1 = -1$

$3 \times 4 \times 2 = 24$

$-1 + 24 = 23$

$4 - 23 = -19$

$4 \times 0 - 1 \times (-1) = 1$

$1 \times 3 - (-2) \times 0 = 3$

$-2 \times (1) - 4 \times 3 = 2 - 12 = -10$

$-1 \times 2 - 0 \times 0 = -2$

$0 \times 3 - 3 \times 2 = -6$

$3 \times 0 - (-1) \times 1 = 1$

$0 \times 1 - 2 \times 4 = -8$

$2 \times (-2) - 1 \times 1 = -1 - 7 = -5$

$1 \times 4 - 0 \times (-2) = 4$

$$\begin{pmatrix} 1 & 3 & -10 \\ -2 & -6 & 1 \\ -8 & -5 & 4 \end{pmatrix}$$

4. Método de Eliminação de Gauss

Utilize o Método de Eliminação de Gauss para resolver o sistema:

$$\begin{cases} 2x + y - z = 5 \\ x - 3y + 2z = -1 \\ 3x + 2y + z = 6 \end{cases}$$

Valor: 2,0 pontos

$$\left[\begin{matrix} 2 & 1 & -1 \\ 1 & -3 & 2 \\ 3 & 2 & 1 \end{matrix}\middle|\begin{matrix} 5 \\ -1 \\ 6 \end{matrix}\right] \quad \left[\begin{matrix} 1 & -3 & 2 \\ 2 & 1 & -1 \\ 3 & 2 & 1 \end{matrix}\middle|\begin{matrix} -1 \\ 5 \\ 6 \end{matrix}\right] \rightarrow \left[\begin{matrix} 1 & -3 & 2 \\ 0 & 7 & -5 \\ 3 & 2 & 1 \end{matrix}\middle|\begin{matrix} -1 \\ 7 \\ 6 \end{matrix}\right] \rightarrow \left[\begin{matrix} 1 & -3 & 2 \\ 0 & 7 & -5 \\ 0 & 11 & -5 \end{matrix}\middle|\begin{matrix} -1 \\ 7 \\ 9 \end{matrix}\right] \rightarrow \left[\begin{matrix} 1 & -3 & 2 \\ 0 & 7 & -5 \\ 0 & 0 & \frac{20}{7} \end{matrix}\middle|\begin{matrix} -1 \\ 7 \\ -2 \end{matrix}\right]$$

$L2 \leftrightarrow L1$

$L2: L2 - \frac{2}{1} \times L1$

$2 - 2 \times 1 = 0$

$1 - 2 \times (-3) = 1 + 6 = 7$

$-1 - 2 \times 2 = -1 - 4 = -5$

$5 - 2 \times (-1) = 5 + 2 = 7$

$L3: L3 - \frac{3}{1} \times L1$

$3 - 3 \times 1 = 0$

$2 - 3 \times (-3) = 2 + 9 = 11$

$1 - 3 \times 2 = 1 - 6 = -5$

$6 - 3 \times (-1) = 6 + 3 = 9$

$L3: L3 - \frac{11}{7} \times L2$

$11 - \frac{11}{7} \times 7 = 11 - \frac{77}{7} = \frac{77}{7} - \frac{77}{7} = 0$

$-5 - \frac{11}{7} \times (-5) = -5 + \frac{55}{7} = \frac{-35}{7} + \frac{55}{7} = \frac{20}{7}$

$9 - \frac{11}{7} \times 7 = -2$

$$\begin{cases} 1x - 3y + 2z = -1 \\ 7y - 5z = 7 \\ \frac{20}{7}z = -2 \end{cases}$$

$\frac{20}{7} z = -2$

$z = \frac{7}{20} \times \frac{-2}{1} = \frac{-14}{20}$

$z = -\frac{7}{10}$

$7y - 5 \times \frac{7}{10} = 7$

$7y - \frac{35}{10} = 7$

$7y = 7 - \frac{35}{10}$

$7y = \frac{70}{10} - \frac{35}{10}$

$7y = \frac{35}{10}$

$y = \frac{35}{10} \times \frac{1}{7}$

$y = \frac{35}{70} = \frac{1}{2}$

$y = \frac{1}{2}$

$1x - 3 \times \frac{1}{2} + 2 \times \left(-\frac{7}{10}\right) = -1$

$1x - \frac{3}{2} + \left(-\frac{14}{10}\right) = -1$

$1x - \frac{3}{2} + \left(-\frac{7}{5}\right) = -1$

$1x - \frac{15}{10} + \frac{-14}{10} = -1$

$1x - \frac{29}{10} = -1$

$1x = -1 + \frac{29}{10}$

$x = \frac{-10}{10} + \frac{29}{10} = \frac{19}{10}$

$x = \frac{19}{10}$

5. Transformação Linear

Sabendo que $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ é uma transformação linear e que

$$T(1,1) = (4,0), \quad T(2,-1) = (1,3)$$

Determine $T(x,y)$.

Valor: 2,0 pontos

$$a\begin{bmatrix} 1 \\ 1 \end{bmatrix} + b\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$1a + 2b = x$

$1a + (-1)b = y \Leftrightarrow 1a - 1b = y$

$(1a + 2b) + (1a - 1b) = x + y$

$2a + 1b = x + y$

$b = x + y - 2a$

$1a + 2(x + y - 2a) = x$

$1a + 2x + 2y - 4a = x$

$-3a + 2x + 2y = x$

$-3a + 2y = x - 2x$

$-3a = -1x - 2y$

$a = \frac{-1x - 2y}{-3}$

$a = \frac{1x + 2y}{3}$

$b = x + y - 2\left(\frac{1x + 2y}{3}\right)$

$b = x + y - \frac{2x - 4y}{3}$

$b = \frac{3x + 3y}{3} - \frac{2x - 4y}{3}$

$b = \frac{3x + 3y - 2x - 4y}{3}$

$b = \frac{1x - 1y}{3}$

$a = \frac{1x + 2y}{3} \qquad b = \frac{1x - 1y}{3}$

$$\frac{1x + 2y}{3}\begin{bmatrix} 4 \\ 0 \end{bmatrix} + \frac{1x - 1y}{3}\begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

1ª coordenada

$\frac{1x + 2y}{3} \times 4 + \frac{1x - 1y}{3} \times 1$

$\frac{4x + 8y}{3} + \frac{1x - 1y}{3} \Leftrightarrow$

$\frac{5x + 7y}{3}$

2ª coordenada

$\frac{1x + 2y}{3} \times 0 + \frac{1x - 1y}{3} \times 3 \Leftrightarrow$

$\frac{1x - 1y}{3} \times 3 \Leftrightarrow$

$x - y$

$$T(x,y) = \left(\frac{5x + 7y}{3}, \; x - y\right)$$

$T(1,1) = \left(\frac{5 \times 1 + 7 \times 1}{3}, \; 1 - 1\right) = \left(\frac{5 + 7}{3}, 0\right) = \left(\frac{12}{3}, 0\right) = (4, 0)$ ✓

$T(2,-1) = \left(\frac{5 \times 2 + 7 \times (-1)}{3}, \; 2 + 1\right) = \left(\frac{10 - 7}{3}, 3\right) = \left(\frac{3}{3}, 3\right) = (1, 3)$ ✓

6. Independência Linear

Verifique se os vetores

$$u_1 = (1, -1, 2), \quad u_2 = (3, 0, 1), \quad u_3 = (0, 2, -1)$$

são linearmente independentes em $\mathbb{R}^3$.

$$\begin{matrix} 1 & 3 & 0 \\ -1 & 0 & 2 \\ 2 & 1 & -1 \end{matrix} \left| \begin{matrix} 1 & 3 \\ -1 & 0 \\ 2 & 1 \end{matrix} \right.$$

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$3 \times 2 \times 2 = 12$

12

$3 \times (-1) \times (-1) = 3$

$1 \times 2 \times 1 = 2$

$3 + 2 = 5$

$12 - 5 = 7$